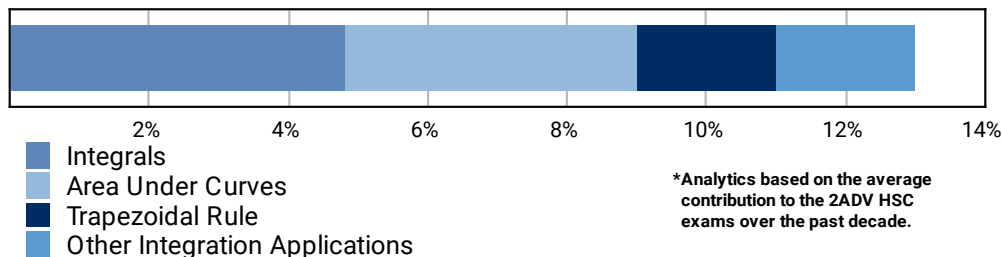


ADV: Calculus (Adv), C4 Integration (Adv) Other Integration Applications (Y12)

Teacher: Denis Vlismas

Exam Equivalent Time: 52.5 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

C4 Integration



HISTORICAL CONTRIBUTION

- C4 Integration* has contributed a substantial 13.0% to past Advanced exams, on average over the past decade.
- This topic has been split into four sub-topics for analysis purposes: 1-Integrals (4.8%), 2-Areas Under Curves (4.2%), 3-Trapezoidal Rule (2.0%) and 4-Other Integration Examples (2.0%).
- This analysis looks at *Other Integration Applications*.

HSC ANALYSIS - What to expect and common pitfalls

- Other Integration Applications* was examined in 2021 in a significant 8-mark question that combined the cross topic of trig graphs. The potential importance of this area was originally flagged by a 6-mark allocation in NESA's sample Advanced exam and is now confirmed.
- Students need a good understanding of calculating quantities from the integration of rate of change equations and graphs.
- Some *EQ-Bank* questions in the database take the form of shorter questions that we expect to feature in the new exam format, consistent with NESA's sample exam.
- Underlying themes can be very broad, 2021 Q27 looked at energy charge, and other database examples deal with motion equations and exponential growth equations (see 2ADV *EQ-Bank* 2 and 5).

Questions

1. Calculus, 2ADV C4 SM-Bank 1 MC

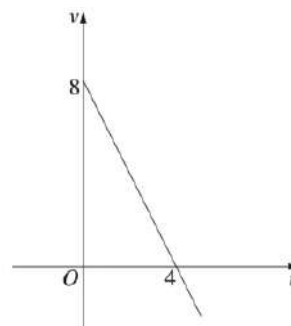
A lift accelerates from rest at a constant rate until it reaches a speed of 3 ms^{-1} . It continues at this speed for 10 seconds and then decelerates at a constant rate before coming to rest. The total travel time for the lift is 30 seconds.

The total distance, in metres, travelled by the lift is

- A. 45
- B. 60
- C. 75
- D. 90

2. Calculus, 2ADV C4 2015 HSC 9 MC

A particle is moving along the x -axis. The graph shows its velocity v metres per second at time t seconds.



When $t = 0$ the displacement x is equal to 2 metres.

What is the maximum value of the displacement x ?

- (A) 8 m
- (B) 14 m
- (C) 16 m
- (D) 18 m

3. Calculus, 2ADV C4 EQ-Bank 2

The population, D , of Tasmanian Devils in a sanctuary is given by $D(t)$, where t is the time in years after the sanctuary was established.

The devil population changes at a rate modelled by the function $\frac{dD}{dt} = 28e^{0.35t}$.

Calculate the increase in the number of Tasmanian Devils at the end of the first 8 years. Give your answer correct to three significant figures. (3 marks)

4. Calculus, 2ADV C4 2016 HSC 16a

A particle moves in a straight line. Its velocity $v \text{ ms}^{-1}$ at time t seconds is given by

$$v = 2 - \frac{4}{t+1}.$$

- Find the initial velocity. (1 mark)
 - Find the acceleration of the particle when the particle is stationary. (2 marks)
 - By considering the behaviour of v for large t , sketch a graph of v against t for $t \geq 0$, showing any intercepts. (2 marks)
 - Find the exact distance travelled by the particle in the first 7 seconds. (3 marks)
-

5. Calculus, 2ADV C4 2012 HSC 15b

The velocity of a particle is given by

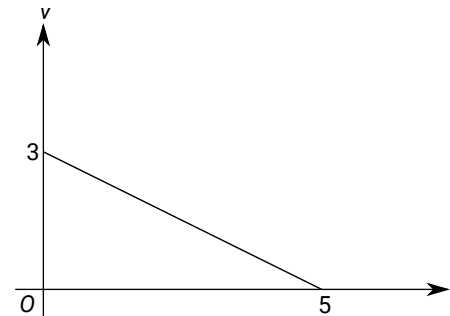
$$v = 1 - 2 \cos t,$$

where x is the displacement in metres and t is the time in seconds. Initially the particle is 3 m to the right of the origin.

- Find the initial velocity of the particle. (1 mark)
 - Find the maximum velocity of the particle. (1 mark)
 - Find the displacement, x , of the particle in terms of t . (2 marks)
 - Find the position of the particle when it is at rest for the first time. (2 marks)
-

6. Calculus, 2ADV C4 EQ-Bank 5

The velocity of a particle moving along the x -axis at v metres per second at t seconds, is shown in the graph below.



Initially, the displacement x is equal to 12 metres.

- Write an equation that describes the displacement, x , at time t seconds. (2 marks)
 - Draw a graph that shows the displacement of the particle, x metres from the origin, at a time t seconds between $t = 0$ and $t = 5$. Label the coordinates of the endpoints of your graph. (2 marks)
-

7. Calculus, 2ADV C4 2021 HSC 27

Kenzo has a solar powered phone charger. Its power, P , can be modelled by the function

$$P(t) = 400 \sin\left(\frac{\pi}{12}t\right), \quad 0 \leq t \leq 12,$$

where t is the number of hours after sunrise.

a. Sketch the graph of P for $0 \leq t \leq 12$. (2 marks)

Power is the rate of change of energy. Hence the amount of energy, E units, generated by the solar powered phone charger from $t = a$ to $t = b$, where $0 \leq a \leq b \leq 12$ is given by

$$E = \int_a^b P(t) dt.$$

b. Show that $E = \frac{4800}{\pi} \left(\cos \frac{a\pi}{12} - \cos \frac{b\pi}{12} \right)$. (2 marks)

c. To make a phone call, a phone battery needs at least 300 units of energy. Kenzo woke up 3 hours after sunrise and found that his phone battery had no units of energy. He immediately began to use his solar powered charger to charge his phone battery.

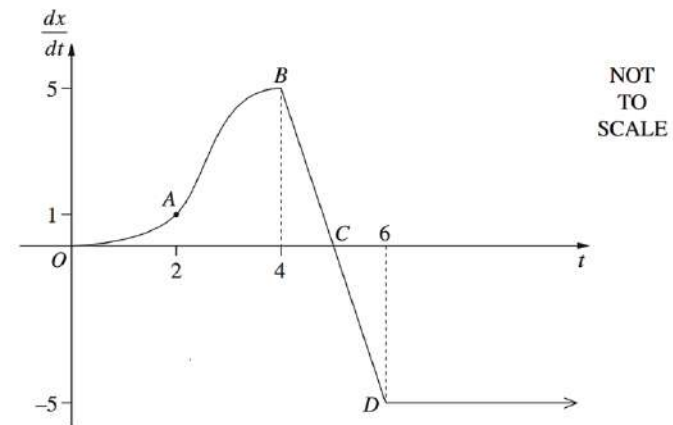
Find the least amount of time he needed to wait before he could make a phone call. Give your answer correct to the nearest minute. (3 marks)

d. The next day, Kenzo woke up 6 hours after sunrise and again found that his phone battery had no units of energy. He immediately began to use his solar powered charger to charge his phone battery.

Would it take more time or less time or the same amount of time, compared to the answer in part (c), to charge his phone battery in order to make a phone call? Explain your answer by referring to the graph drawn in part (a). (1 mark)

8. Calculus, 2ADV C4 2007* HSC 10a

An object is moving on the x -axis. The graph shows the velocity, $\frac{dx}{dt}$, of the object, as a function of time, t . The coordinates of the points shown on the graph are $A(2, 1)$, $B(4, 5)$, $C(5, 0)$ and $D(6, -5)$. The velocity is constant for $t \geq 6$.

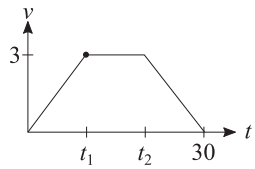


- The object is initially at the origin. During which time(s) is the displacement of the object decreasing? (1 mark)
- If the object travels 7 units in the first 4 seconds, estimate the time at which the object returns to the origin. Justify your answer. (2 marks)
- Sketch the displacement, x , as a function of time. (2 marks)

Worked Solutions

1. Calculus, 2ADV C4 SM-Bank 1 MC

Consider the velocity graph:



$$t_1 \rightarrow t_2 = 10 \text{ seconds}$$

$$\begin{aligned}\therefore \text{Total distance travelled} &= \text{Area of trapezium} \\ &= \frac{1}{2} \times 3(10 + 30) \\ &= 60 \text{ m}\end{aligned}$$

$\Rightarrow B$

2. Calculus, 2ADV C4 2015 HSC 9 MC

Distance travelled

$$\begin{aligned}&= \int_0^4 v \, dt \\ &= \text{Area under the velocity curve} \\ &= \frac{1}{2} \times b \times h \\ &= \frac{1}{2} \times 4 \times 8 \\ &= 16 \text{ metres.}\end{aligned}$$

Since velocity is always positive between $t = 0$ and $t = 4$, and the original displacement = 2, the maximum displacement = $16 + 2 = 18$ metres
 $\Rightarrow D$

3. Calculus, 2ADV C4 EQ-Bank 2

$$\begin{aligned}\int_0^8 28e^{0.35t} &= \left[28 \times \frac{1}{0.35} e^{0.35t} \right]_0^8 \\ &= 80(e^{0.35 \times 8} - e^0) \\ &= 80(16.44... - 1) \\ &= 1235.57... \\ &= 1240 \text{ (to 3 sig. fig.)}\end{aligned}$$

4. Calculus, 2ADV C4 2016 HSC 16a

i. Initial velocity when $t = 0$

$$v = 2 - \frac{4}{1} = -2 \text{ ms}^{-1}$$

Mean mark 93%.

COMMENT: Great example of low hanging fruit late in the exam for students who progress efficiently.

ii. $v = 2 - \frac{4}{t+1}$

$$a = \frac{dv}{dt} = \frac{4}{(t+1)^2}$$

Particle is stationary when $v = 0$,

$$2 - \frac{4}{t+1} = 0$$

$$2(t+1) = 4$$

$$t = 1$$

When $t = 1$,

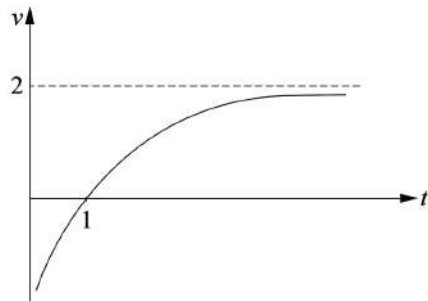
$$\therefore a = \frac{4}{(1+1)^2}$$

$$= 1 \text{ ms}^{-2}$$

iii. $v = 2 - \frac{4}{t+1}$

As $t \rightarrow \infty$, $\frac{4}{t+1} \rightarrow 0$

$$\therefore v \rightarrow 2$$



iv. Distance travelled in 1st 7 seconds

♦♦ Mean mark 26%.

$$= \left| \int_0^1 \left(2 - \frac{4}{t+1} \right) dt \right| + \int_1^7 \left(2 - \frac{4}{t+1} \right) dt$$

$$= -[2t - 4 \ln(t+1)]_0^1 + [2t - 4 \ln(t+1)]_1^7$$

$$= -[(2 - 4 \ln 2) - 0] + [(14 - 4 \ln 8) - (2 - 4 \ln 2)]$$

$$= 4 \ln 2 - 2 + 12 - 4 \ln 2^3 + 4 \ln 2$$

$$= 10 + 8 \ln 2 - 12 \ln 2$$

$$= 10 - 4 \ln 2 \text{ metres}$$

♦ Mean mark 47%.

5. Calculus, 2ADV C4 2012 HSC 15b

i. Find v when $t = 0$:

$$\begin{aligned} v &= 1 - 2 \cos 0 \\ &= 1 - 2 \\ &= -1 \end{aligned}$$

$\therefore$ Initial velocity is -1 m/s.

ii. Solution 1

Max velocity occurs when $a = \frac{dv}{dt} = 0$

$$a = 2 \sin t$$

Find t when $a = 0$:

$$\begin{aligned} 2 \sin t &= 0 \\ t &= 0, \pi, 2\pi, \dots \end{aligned}$$

At $t = 0$, $v = -1$ m/s

At $t = \pi$, $v = 1 - 2(-1) = 3$ m/s

$\therefore$ Maximum velocity is 3 m/s

Solution 2

$$v = 1 - 2 \cos t$$

Since $-1 < \cos t < 1$

$$-2 < 2 \cos t < 2$$

$$-1 < 1 - 2 \cos t < 3$$

$\therefore$ Maximum velocity is 3 m/s

iii. $x = \int v \, dt$

$$= \int (1 - 2 \cos t) \, dt$$

$$= t - 2 \sin t + c$$

When $t = 0$, $x = 3$ (given)

$$3 = 0 - 2 \sin 0 + c$$

$$c = 3$$

$$\therefore x = t - 2 \sin t + 3$$

iv. Find x when $v = 0$ (first time):

When $v = 0$,

$$0 = 1 - 2 \cos t$$

$$\cos t = \frac{1}{2}$$

$$t = \cos^{-1}\left(\frac{1}{2}\right)$$

$$= \frac{\pi}{3} \quad (\text{first time})$$

Find x when $t = \frac{\pi}{3}$:

$$x = \frac{\pi}{3} - 2 \sin\left(\frac{\pi}{3}\right) + 3$$

$$= \frac{\pi}{3} - 2 \times \frac{\sqrt{3}}{2} + 3$$

$$= \frac{\pi}{3} - \sqrt{3} + 3 \quad \text{units}$$

♦♦ Mean mark 29%

MARKER'S COMMENT: Solution 2 is more efficient here. Using the -1 and +1 limits of trig functions can be very a effective way to calculate max/min values.

♦ Mean mark 50%

MARKER'S COMMENT: Many students found $t = \frac{\pi}{3}$ but failed to gain full marks by omitting to find x . Remember that for calculus, angles are measured in radians, NOT degrees!

6. Calculus, 2ADV C4 EQ-Bank 5

$$\text{i. } m_v = -\frac{3}{5}$$

$$v = 3 - \frac{3}{5}t$$

$$x = \int v \, dt$$

$$= \int 3 - \frac{3}{5}t \, dt$$

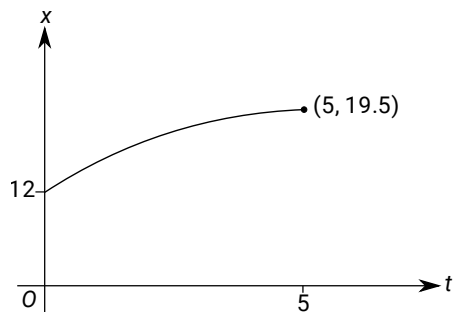
$$= 3t - \frac{3}{10}t^2 + c$$

When $t = 0, x = 12 \Rightarrow c = 12$

$$\therefore x = 3t - \frac{3}{10}t^2 + 12$$

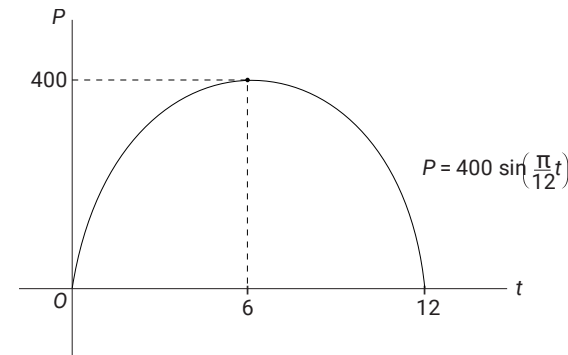
ii. When $t = 5$:

$$x = 15 - \frac{3}{10} \times 25 + 12 = 19.5$$



7. Calculus, 2ADV C4 2021 HSC 27

a.



$$\text{b. } E = \int_a^b 400 \sin\left(\frac{\pi}{12}t\right) dt$$

$$= \left[-400 \cdot \frac{12}{\pi} \cos\left(\frac{\pi}{12}t\right) \right]_a^b$$

$$= -\frac{4800}{\pi} \cos\left(\frac{\pi}{12}b\right) - \left(-\frac{4800}{\pi} \cos\left(\frac{\pi}{12}a\right) \right)$$

$$= \frac{4800}{\pi} \left(\cos \frac{a\pi}{12} - \cos \frac{b\pi}{12} \right)$$

c. Find b given $E = 300$ and $a = 3$:

$$300 = \frac{4800}{\pi} \left(\cos \frac{\pi}{4} - \cos \frac{b\pi}{12} \right)$$

$$\frac{300\pi}{4800} = \frac{1}{\sqrt{2}} - \cos \frac{b\pi}{12}$$

$$\cos \frac{b\pi}{12} = \frac{1}{\sqrt{2}} - \frac{\pi}{16}$$

$$\frac{b\pi}{12} = \cos^{-1} \left(\frac{1}{\sqrt{2}} - \frac{\pi}{16} \right)$$

$$b = \frac{12}{\pi} \cos^{-1} \left(\frac{1}{\sqrt{2}} - \frac{\pi}{16} \right)$$

$$= 3.952 \dots$$

$$= 3 \text{ h } 57 \text{ m (nearest minute)}$$

$\therefore$ Least time before phone is charged = 57 minutes

♦♦ Mean mark part (c) 29%
COMMENT: It is arguable that the nearest minute may also be 3h 58 m as the phone is not adequately charged at 3h 57 m.

d. The power produced is at its peak when $t = 6$.

$\therefore$ It will charge in less time.

♦♦ Mean mark part (d) 22%

8. Calculus, 2ADV C4 2007* HSC 10a

i. Displacement is reducing when the velocity is negative.

$\therefore t > 5$ seconds

ii. At B , the displacement = 7 units

Considering displacement from B to D :

Since the area below the graph from

B to C equals the area above the

graph from C to D , there is no change

in displacement from B to D .

Considering $t \geq 6$

Time required to return to origin

$$\begin{aligned} t &= \frac{d}{v} \\ &= \frac{7}{5} \\ &= 1.4 \text{ seconds} \end{aligned}$$

$\therefore$ The particle returns to the origin after 7.4 seconds.

iii.

